Effects of Spontaneous Emission rates on the Fidelity of Quantum Gates

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Abstract

This report first details a simulation made of the CZ, Hadamard and CNOT gates and the optimisation for fidelity by using different shaped Rabi pulses for the laser driven state transitions. The pulse shapes for the Rabi frequency that were simulated were the square pulse, gaussian pulse and a pulse which approximates the rapid adiabatic passage for state transitions. When the system was evolved with unitary operators, the fidelity after the operation was $1 - 10^{-15}$ for the highest fidelity pulse, without decoherence, all pulses returned a fidelity of close to 1. The simulations of step functions were then modified to include the non-unitary evolution that occurs when the system is coupled to the environment in the form of spontaneous emission and the impact this emission had on the gate fidelity. It was found that on the order of a gate time of $3\mu s$ the effects of spontaneous emission were relevant for rates on the order of 100kHz.

1 Introduction

Quantum computing is the forefront of computing technologies. It works with quantum bits, called qubits, instead of bits as with in a classical computer. A bit can take discrete values of 0 or 1, by contrast a qubit can exist as a superposition of states 0 and 1. The number of different states of a n-bit system for a classical computer is 2^n, the quantum system of n-qubits has 2^n mutually orthogonal quantum states.[1] This is the number of pure states of the system, in addition to this, a quantum computer can exist in mixed states. Qubits are formed from very simple quantum mechanical objects, for example an isolated atom with a system of many qubits can be operated on by a single operation to produce a computation. The processing power of classical computers is limited due to the physical number of states that a classical computer can take when compared to a quantum computer. The robustness of quantum computations is still a challenge as the environment has an effect on the system, causing the information stored in the system to be gradually lost over time, called decoherence. These effects due to coupling of the system to the surroundings lead to adiabatic and thermal decoherence as the two main types.[**?**] Fidelity in quantum computing is a measure of the accuracy of a quantum gate in producing the desired result. This is the largest obstacle in quantum computing; how a quantum computer can be made with enough redundancy to produce an accurate computation. As with classical computers, there are bits stored to pick up the slack when there is an accidental bit flip. The bits behave classically and are unlikely to flip, so with enough redundant bits, the computations produced can be trusted. With quantum computers, the qubits will never be in a complete or 'pure' state once a computation has been made, and the decoherence will occur to all qubits in the system as it is a property of the environment[2] Quantum gate simulations allow us to model a quantum operation on a system of qubits. The most common quantum gates are CZ, Hadamard and CNOT, these can be implemented in combinations to give a specific computation. The physical system is able to be driven by an applied laser pulse to different energy levels, which amounts to having a qubit change value. The physical evolution of the system can be computed by approximating the driving fields by a Hamiltonian. In the rotating frame, the Hamiltonian of the laser pulse that will be considered is given by Eqn.1,

$$H = \frac{\hbar}{2}\begin{pmatrix} \Delta & \Omega e^{-i\phi} \\ \Omega e^{i\phi} & -\Delta \end{pmatrix} \tag{1}$$

where Δ is the detuning of the laser and is a representation of how close the frequency of the applied laser pulse is to the energy transition between states of the qubit; $\Delta = \omega - \omega_0$, ω is the applied frequency of the laser with ω_0 being the energy associated with the transition of the energy levels, Ω is the applied Rabi frequency and is proportional to the amplitude of the applied laser pulse and ϕ is the phase of the laser. This is applied to the state vector, $|\psi\rangle$, of the system to give the evolution. When the state is evolved, the state vector oscillates sinusoidally, therefore state populations oscillate with a sin^2 relationship.

1.1 STIRAP

STIRAP (stimulated Raman adiabatic passage) is a scheme which has been proposed to give reliable quantum control for state transfer in qubits which utilises an intermediate state$|p\rangle$ for exciting into the Rydberg state. The evolution of the Hamiltonian in this scheme is adiabatic. Modifications to the Hamiltonian such as counterdiabatic driving (CD) have been proposed to reduce the decoherence by speeding up the procedure to reduce time spent in the higher states. The Hamiltonian is formulated as shown in Eqn.2 which includes driving from the $|1\rangle$ to the $|p\rangle$ state in the first term and from the $|p\rangle$ state to the $|r\rangle$ state in the second term.

$$H = \frac{1}{2}[\Omega_p(t)|0\rangle\langle 1| + \Omega_s(t)e^{-i\phi}|1\rangle\langle 2| + h.c][3] \tag{2}$$

This scheme is has been demonstrated to be a method for reducing the effects of decoherence due to the continuous driving of the system, which doesn't allow the system to sit in the excited $|p\rangle$ state.[4] While this report does not deal with the full STIRAP scheme, the Rabi pulse shape that is used in this scheme is considered.

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1.2 Decoherence

Decoherence in quantum computing can come from several sources; for example the applied laser pulse is difficult to get qubits spontaneously emit photons, causing an unwanted relaxation to lower energy states, which is the type of decoherence considered in this investigation. High energy states like the Rydberg state generally have much higher rate of spontaneous emission. This means that a lower time spent in a high energy state needs to be minimised when performing an operation.[5] Spontaneous emission is often mitigated by placing the atom representing a qubit in a cavity, which decreases the coupling to the surroundings.[6] The time evolution of a density matrix can be modified to include the decoherence in the simulation with the addition of a gamma factor as seen in Eqn.3. The solution to this acting upon the state alongside the gate operation is the solution to the Lindblad Master equation.

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H,\rho] + \Gamma \tag{3}$$

The form that Γ takes is dependent on the strength of the coupling of the system to the environment. Eqn.3 can be expanded out to give the total Lindblad master equation for the evolution of the system as in Eqn.4.

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H,\rho] + \sum_i \sqrt{\gamma_i}\left(L_i\rho L_i^\dagger - \frac{1}{2}\left\{L_i^\dagger L_i, \rho\right\}\right) \tag{4}$$

The Lindblad operators L_i describe each type of collapse, these operators are non-unitary which means that information is completely lost from the system. The collapse operator first considered was that for spontaneous emission. This is one single collapse operator defined as $L_1 = \sigma_-$[7] and γ_1 is the rate at which spontaneous emission occurs. was adjusted to show strong coupling. The evolution of the density matrix was calculated using the QuTip python package for the Lindblad master equation solver.

2 Methods

The simulations in this method are approximations of a two or three level quantum system, existing on separate energy levels, where a laser of specific Rabi frequency and detuning is used to excite each qubit. Both the Rabi frequency and the detuning are implemented as time dependent pulses and sweeps, meaning that the time evolution of the state needed to be "Trotterised" ie. applied in small separate time steps as time independent Hamiltonians, according to Eqn.5.

$$|\psi(t+\delta t)\rangle = e^{-i\hbar H(t)\delta t}|\psi(t)\rangle \tag{5}$$

The parameter τ in Eqn.17 is equal to the integral of the rabi frequency over the pulse duration. This variable decides the transition that occurs on the qubit which amounts to a specific rotation on the Bloch sphere about a certain axis. Throughout this investigation $\hbar$ was set to 1, since the specific energy of the state was not relevant.
The three pulse shapes that were considered were the idealise square pulse which is not possible in practice as it requires a step change in amplitude. The two 'realistic' pulse shapes were the gaussian pulse and a pulse represented by Eqn.6, where τ is an arbitrary width associated with the pulse, Ω_0 being the maximum Rabi frequency, and N being a normalisation constant. The two pulses are shown in Fig.1, (a) is proposed to be more effective than the gaussian as the sides are steeper, allowing a shorter pulse width and a quicker excitation.

$$\Omega(t) = \frac{\Omega_0}{N}\left(exp\left(-\frac{(2t-\tau)^2}{4\tau^2}\right) + 2t(2t-\tau)\right) \tag{6}$$

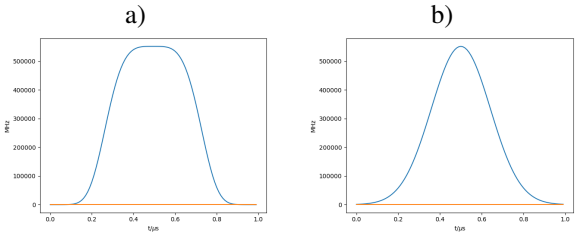


Figure 1: a) representation of Eqn.6 for an adiabatic rapid passage as a function of time b) the gaussian pulse, both applied to give the CZ gate

The other, effectively identical way to evolve the state is to apply the unitary transformation to the density matrix.

$$\rho = |\psi\rangle\langle\psi| \tag{7}$$

$$\rho(t) = U\rho(0)U^\dagger \tag{8}$$

Where the unitary transformation U takes the same form as with the state vector representation and $U^\dagger$ is the Hermitian conjugate.
The measure of how well a quantum gate gives the correct output is called fidelity. The ideal gate is represented as a matrix, and is used as the operation to compare against when calculating fidelity. The metric used was the comparison of the output density matrices of the ideal and the simulated gate. This is calculated as

$$F(\rho_2,\rho_1) = Tr(\rho_2\rho_1) \tag{9}$$

where ρ_1 is the target state and ρ_2 is the simulated state, this method only relies upon the fidelity calculation from a single input state rather than averaging over many different inputs, which makes it less reliable. [8]
The decoherence of a three level system is modelled by a 3x3 collapse operator as

$$L_{coll} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \tag{10}$$

Implementing this into a system of two qubits required selecting the state which the qubit will collapse from for each system state. The highest relaxation rate comes from the $|r\rangle \rightarrow |1\rangle$ collapse which was represented by the collapse operator $\sqrt{\gamma}L_{coll}$ where L_{coll} is a 9x9 matrix and has 1s in positions $L_{23}, L_{47}, L_{56}, L_{58}, L_{69}, L_{89}$. The lesser relaxation rate comes from the $|1\rangle \rightarrow |0\rangle$ channel, which is a matrix of the same dimension but with 1s in positions $L_{12}, L_{14}, L_{25}, L_{36}, L_{45}, L_{78}$.

3 CZ gate

The CZ (or controlled-Z) gate operates on a system of two qubits. The effect of this gate is to change the sign of the doubly excited $|11\rangle$ state. The matrix representation shown as a matrix of rank 4 in Eqn.11 which is applied to the state vector, represented as a complex 4-vector.

$$CZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \tag{11}$$

Each qudit to realise a CZ gate actually needs to exist as a three level system where the second excited state exists at a much higher energy, called a Rydberg state, denoted by $|r\rangle$. The CZ utilises the fact that Rydberg atoms have a property called a Rydberg blockade. In the Rydberg blockade regime, a large 'toll' on the doubly excited state occurs, which causes this state to be rapidly de-excited into lower energy states. The CZ gate works by applying a pulse to the qubits which excites the $|1\rangle$ state into the $|r\rangle$ for both atoms, and sustains this excitation until the $|11\rangle$ accumulates a phase of π due to the strong interaction which results in a sign change. This only works when the interaction strength $V >> \Omega$. The targeted excitation for the $|1\rangle \rightarrow |r\rangle$ excitation Hamiltonian is given by Eqn.12 where Δ is the laser detuning. In this representation we have kept the detuning Δ at zero as well as the phase ϕ. This is an idealistic representation if the Hamiltonian since it assumes the $|0\rangle \rightarrow |1\rangle$ transition requires a much higher energy so that $\Delta_{0,1} = |\omega - \omega_{0,1}| >> 0$ which means that the laser isn't resonating with this state transition but is with the $|1\rangle \rightarrow |r\rangle$ transition. The atoms we are using here are assumed to be identical since the Hamiltonians being applied are global and affect both qubits.[9]

$$H = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \Delta & \Omega e^{i\phi} \\ 0 & \Omega e^{i\phi} & -\Delta \end{pmatrix} \tag{12}$$

The Hamiltonian is formulated as shown in Eqn.13, which shows the individual Hamiltonians applied to the control qubit, H_c and target qubit H_t

$$H = H_c \otimes I + I \otimes H_t + V|rr\rangle\langle rr| \tag{13}$$

V is the interaction potential between qubits in the Rydberg state. This interaction strength is a property of the atoms and their proximity to each other given by Eqn.14

$$\hbar V = \frac{C_6}{R^6} \tag{14}$$

where C_6 is a constant which increases with principal quantum number and R is the separation. The value of the Rabi frequency was determined by using Fig.2 to find the value for which the fidelity was at a maximum.

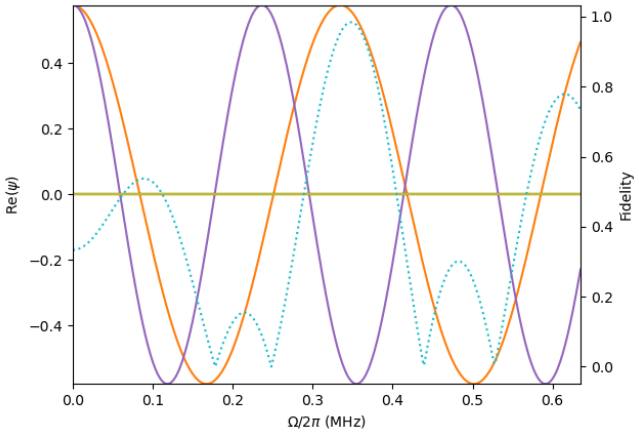


Figure 2: The real component of the state vector components output from the CZ gate, shown as the solid lines, and the total gate fidelity shown as the dotted blue line, when varying the Rabi frequency. Detuning of the system was kept at zero and the shape of the Rabi pulse is square, the Hamiltonian was applied over a period of $3\mu s$. The purple line shows the $|11\rangle$ state, the red line shows the $|01\rangle$ and $|10\rangle$ states and the yellow line shows all other states. The input state was $|\psi\rangle = |00\rangle + |01\rangle + |11\rangle$

As seen by Fig.2, there is a mismatch in the time that the $|11\rangle$ state reaches a minimum and the time that the $|10\rangle$ and $|01\rangle$ states reach a maximum. This is a certain property of the type of pulse we are applying so the gate cannot be simulated with a fidelity of exactly 1.
Fig.3 demonstrates how the populations oscillate between $|01\rangle$ and $|0r\rangle$ and $|10\rangle$ and $|r0\rangle$ states, where the state vector with either of the qubits in the $|0\rangle$ state are evolved for one period, whereas the state vector of the state we want to accumulate a phase of π has been evolved for 1.5 periods. The $|00\rangle$ state stays constant as the only transition being implemented is the $|1\rangle \rightarrow |r\rangle$.

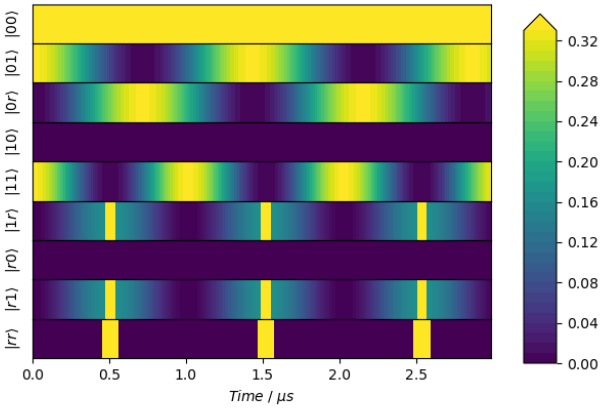


Figure 3: The evolution of populations of states state as a function of time for a CZ gate acting on an initial state of $|\psi\rangle = |00\rangle + |01\rangle + |10\rangle + |11\rangle$ and a constant detuning and Rabi frequency of $\Delta/2\pi = 0$ and $\Omega/2\pi = 0.3482MHz$. The solid yellow lines in the $|1r\rangle$, $|r1\rangle$ and $|rr\rangle$ states are a result of the strong Rydberg blockade on the $|rr\rangle$ state.

4 CNOT gate

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{15}$$

The purpose of the CNOT gate is to flip the sign of the target qubit when the control qubit is in the excited state. The CNOT gate in practice is formed of a CZ gate which has been transformed into the Hadamard basis by a Hadamard gate. The state is transformed back into the original basis by another Hadamard gate. The state vector components of states $|10\rangle$ and $|11\rangle$ swap after the gate has been applied, which is seen as a swap of their probability in Fig.4. [10]

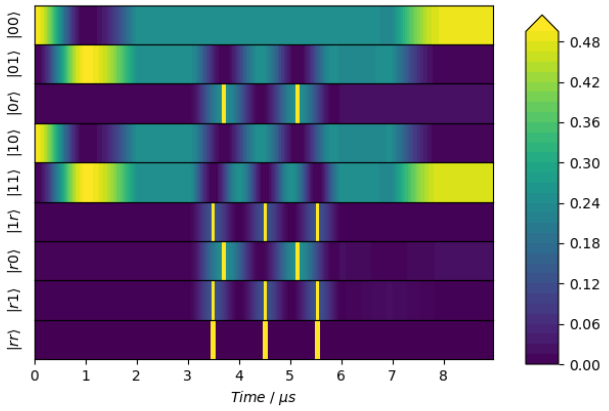


Figure 4: The evolution of the state populations of the system as a function of time for a CNOT gate, shown by the colour intensity. The input state being $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$. The gate was realised with a three pulse evolution, with the middle section being the CZ gate and the Hadamard gates either side. The detuning was kept at zero for the CZ gate and $\Omega/2\pi$ was set to 0.32MHz for a pulse duration of $9.17\mu s$

4.1 Hadamard Gate

The Hadamard gate changes the basis of the qubits states to the Hadamard basis, effectively creating a maximally entangled state from a pure state and a pure state from a mixed state, called a Bell state. Repeat application of the Hadamard gate returns the system to the original state.

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{16}$$

It is realised by a series of three rotations on the Bloch sphere, where all rotations are unitary operations, conserving probability. A rotation of π about the x-axis followed by a rotation of $-\pi/2$ about the y-axis, then a final operation which gives a phase shift of π, giving the correct real and imaginary components of the output state vector. The first operation on the single qubit is done using Eqn.1 as the Hamiltonian.

$$\left|\begin{array}{ll} \text{x-axis rotation} & H_x = \begin{pmatrix} 0 & \Omega(t) \\ \Omega(t) & 0 \end{pmatrix} \\ \text{y-axis rotation} & H_y = \begin{pmatrix} 0 & \Omega(t)e^{i\pi/2} \\ \Omega(t)e^{-i\pi/2} & 0 \end{pmatrix} \end{array}\right.$$

The Rabi frequency is applied so that the integration over the time of the pulse gives the correct rotation, as given by Eqn.17. The evolution is then applied to the qubit in the form of Trotter-Suzuki decomposition as in Eqn.5.

$$\int_0^T \Omega(t)dt = -\theta/2 \tag{17}$$

This gives control over the pulse shape since the gate can be realised as long as this integration is correct. The evolution is continued with the rotation about the y-axis with the rabi frequency applied being phase shifted by $\phi = \pi/2$. The final evolution doesn't change the physical state of the system so the Rabi frequency is turned off and the evolution is modelled with the identity matrix multiplied by the Rabi frequency for a rotation of π as the Hamiltonian.[10]

4.2 Realising CNOT

The Hadamard gate is only applied only to the target qubit in this simulation. In practice, this requires the atoms to be separated from each other in this step in order for the laser pulse to only affect one qubit.

This local pulse is represented as a 9x9 matrix for the whole system, with zero components so that only the $|0\rangle \rightarrow |1\rangle$states of the target qubit are influenced as Eqn.18

$$H_{Hadamard} = \begin{pmatrix} \Delta & \Omega e^{i\phi} & 0 \\ \Omega e^{-i\phi} & -\Delta & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{18}$$

The total Hamiltonian calculated by taking the tensor product with the identity matrix to give a Hamiltonian for the total system as Eqn.19

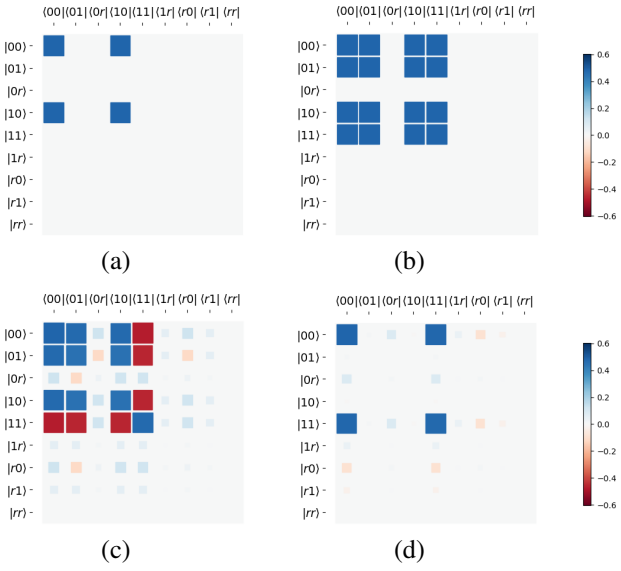


Figure 5: The evolution of the density matrix for the CNOT gate at the end of each step. a) the initial state b) the state after the first Hadamard has been applied c) the state after the CZ gate has been applied to the state in the Hadamard basis d) the final state after it has been transformed back to the original basis. The Rabi pulses used were idealised square, applied for $1\mu s$ for $a \rightarrow b$ and $c \rightarrow d$ and $3\mu s$ for $b \rightarrow c$.

$$H = I \otimes H_{Hadamard} \tag{19}$$

The Hadamard gate is able to act upon either the target or control qubit for this gate to be realised.

Fig.4 shows the simulated evolution of the state populations for a CNOT gate acting on an initial state of $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$ which returns a state where the populations of the $|10\rangle$ and the $|11\rangle$ states have been inverted. Some information has been left behind in other components and complex values over the evolution of the system. This is best seen in Fig.5 where the smaller squares in the final state represent the loss in fidelity of the gate as it is evolved.

5 Results

The first case to realise the first case of the Hadamard gate was using a step function for the Rabi frequency for each rotation and a detuning of zero. For a pulse width of $1\mu s$, the square Rabi pulses had had an amplitude of $\Omega/2\pi = 0.25MHz$ for the x-axis rotation and an amplitude of $\Omega/2\pi = 0.125MHz$ for the y-axis rotation which gave a gate fidelity of 0.999876632481655. The second pulse applied was a gaussian, where each rotation was applied as a separate gaussian pulse. This gave a fidelity of 0.999999958205041 which turned out to be greater than the step function. The third pulse was a function which is typically used for an adiabatic rapid passage as seen in Eqn.6 and Fig.1(a)[4]. This gave a Hadamard gate fidelity of $1-10^{-15}$. The CZ gate was also approached in the same way, by utilising different Rabi pulse shapes to determine the effect it has on the fidelity of the gate. The single valued function gave a gate fidelity of 0.9699551376133968, for a gate length of $3\mu s$, $\Omega/2\pi = 0.3482MHz$

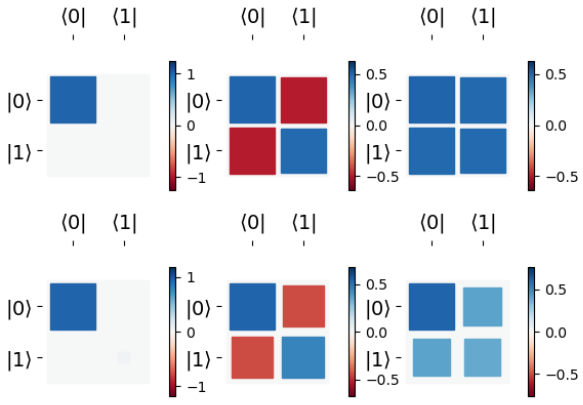
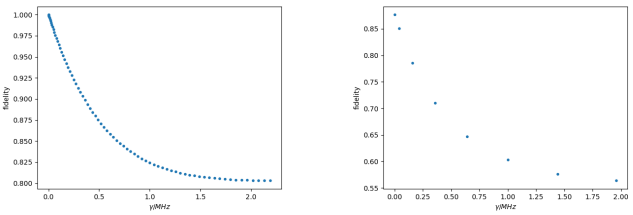


Figure 6: The evolution of the density matrix for a Hadamard gate when there is no interaction with the environment (top) and (bottom) when there is decoherence due to spontaneous emission, using the collapse operator $L = \sigma_-$. The Hamiltonian for both was equal and the Rabi frequency followed a step function. The first column shows the input state, the second shows the state rotated about the x-axis, the third shows the final state. Each step occurs in $1\mu s$, γ was adjusted to give strong coupling to the environment



(a) (b)

Figure 7: A plot of fidelity against gamma factor for (a) a Hadamard gate and (b) a CZ gate. For the Hadamard gate, the fidelity was averaged over three different input states, while for the CZ gate, the fidelity was averaged over two. the shape of the Rabi pulses were square waves. The gate time was $3\mu s$.

6 Discussion

The implementation of the gates is only an approximation, firstly because the time evolution of the state vector has been Trotterised and there's an inherent error associated with that type of generalisation. Secondly this simulation relies upon the rotating wave approximation for the Hamiltonian formulation. One large source of error for the fidelity calculation was that the value for different input states varied a lot due to the length of time each one was left in the excited state for, and an average needed to be calculated across many different inputs. The time taken for the simulation increased for added input states especially when the collapse operators were introduced, so only a small set of inputs could be used. In the selection of the optimal Rabi frequency for the CZ gate, only an observation could be made to read off the maximum fidelity from the graph as the relationship between fidelity and Rabi frequency could not be modelled. The maximum fidelity for the CZ gate was lower than that for the Hadamard gate, due to the mismatch between the $|11\rangle$ state reaching a minimum and the other states reaching a maximum in the CZ gate. This could potentially be reduced by evolving the system for more oscillations to find a point where these extrema line up more, however due to decoherence rates, in practice, this would lead to more information loss and therefore an overall reduction in fidelity. The input states took the form of either pure or mixed states, since the CZ and CNOT gates produced the correct output with mixed states and mixed states were needed for improved fidelity calculations.

The effect that spontaneous emission has on the final state of the system is best shown in Fig.6 where the more highly excited state, $|1\rangle$ has lost some of its information compared to $|0\rangle$. Fig.7 shows how the loss of information affects the overall fidelity of the quantum gates. The curve becomes flat when the operation no longer resembles the gate - effectively the fidelity of a random state with the expected output. The relationship seen is similar for both the Hadamard and CZ gate, suggesting that the loss in fidelity for the CNOT gate would follow a similar trend, much fewer points were able to be acquired for the CZ gate as the computation was much longer. The gamma factor for the spontaneous emission is inversely proportional to the rate at which the excited state relaxes into the lower energies. This is higher for more highly excited states, as in the Rydberg states considered in

this investigation. More highly excited states are required to give the interaction potential essential for the operation of the CZ gate, therefore a trade-off for the level of excitation needs to be reached when implementing this gate in practice, depending upon the properties of the quantum system being used. If spontaneous emission is a random process and the loss of information in the system in the form of the sum of state populations can be modelled by exponential decay, a higher relaxation factor would cause the timescale of emission to decrease. This leads to a higher rate of information loss.[2] The decay of the fidelity with gamma factor was not able to be modelled by an exponential decay. A possible explanation for this is that during the gate evolution, the system oscillates between states where the system does not decay when it is in the lower energy state so the decay can no longer be modelled by an exponential. The typical decay rates of a superconducting qubit are on the order of hundreds of kHz[12], which is seen in Fig7 to have an effect on the gate fidelity. This demonstrates that for gate lengths of a few microseconds, the decoherence rates due to spontaneous emission do significantly affect the gate operation. One way this may be mitigated is by using a shorter gate time so that the atoms aren't in the excited state for as long.

This investigation found that the pulse shown in Fig.1 (a) were the most effective, however all fidelities from this investigation were close to 1 when decoherence was not being considered. This suggests that in the idealised regime, the Hadamard gate was able to be realised perfectly, the very slight loss in fidelity coming from the Trotterisation of the evolution. This is in agreement with the model for the integration of the Rabi frequency over the total pulse duration being proportional to the angle that the state is rotated by, which implies that the shape of the pulse is not relevant when decoherence is not considered.
As seen in Fig.7, the average fidelity across multiple input states causes the fidelity of the gate to decrease which would be expected through non-unitary evolution of the state. This relationship was not able to be modelled by an exponential decay. The decay in the CZ gate was seen to be less than that of the Hadamard gate, this is due to the Hadamard gate being applied for a longer time period. For the CNOT gate, Fig.5 clearly shows some residual state vector amplitudes left in states which contribute to the loss in fidelity. As seen by the $a \rightarrow b$ transition, the ideal transformation happens. However for the $b \rightarrow c$ transition, we see a lot of state amplitudes in Rydberg states, which is down to the imperfect representation of the CZ gate component. Then the transformation back we see that, while the Hadamard gate is nearly perfect, some of the population is left in Rydberg states.

7 Conclusion

The CZ and Hadamard gates can be implemented with specific pulses and by varying the shape and height of the pulses can achieve higher fidelities. The CZ was able to be implemented but not with perfect fidelity using idealised pulses which suggests an inherent issue with the formulation of the CZ gate in this simulation. The effect of spontaneous emission contributes to the loss of information in the gate and causes the gate fidelity to decrease on the timescales relevant to the gate operation times. Longer gate sequences would be greatly affected if the spontaneous emission rate is high, which would call for protection of the system from these effects.

References

[1] A. Steane, "Quantum computing" *Rep. Prog. Phys.* **61** 117 (1998), doi: 10.1088/0034-4885/61/2/002

[2] T. D. Ladd, et al. "Quantum Computers", *Nature* **464**, 08812(2010), doi:10.1038/nature08812

[3] W. Zheng, et al., "Optimal control of stimulated Raman adiabatic passage in a superconducting qudit", *npj Quantum Information* **8**, 9 (2022), doi: 10.1038/s41534-022-00521-7

[4] L. Yagüe Bosch, et al.,"Shortcut-to-Adiabatic Controlled-Phase Gate in Rydberg Atoms", *Phys. Rev. Applied* **11**, 034030 (2019), doi:10.1103/PhysRevApplied.11.034030

[5] M. S. Blok, et. al., "Quantum Information Scrambling on a Superconducting Qutrit Processor", *Phys. Rev. X* **11**, 021010 (2021), doi: 10.1103/PhysRevX.11.021010

[6] A. A. Houck, et. al., "Controlling the Spontaneous Emission of a Superconducting Transmon Qubit", *Phys. Rev. Lett.* **101**, 080502 (2008), doi: 10.1103/PhysRevLett.101.080502

[7] C. A. Brasil, F. F. Fanchini, and R. de Jesus Napolitano, "A simple derivation of the Lindblad equation", *Revista Brasileira de Ensino de Física* **35**, 1303 (2013), doi: 10.1590/S1806-11172013000100003

[8] Y. Lu, et. al., "Optimal fidelity estimation for density matrix", *Scientific Reports* **14**, 30335 (2014), doi: 10.1038/s41598-024-82168-2

[9] M. Saffman, et al., "Symmetric Rydberg controlled-Z gates with adiabatic pulses" *Phys. Rev. A* **101**, 062309 (2020), doi: 10.1103/PhysRevA.101.062309

[10] G. Nourallah, S. Cong, "Preparation of the Hadamard and CNOT gates to realize the maximum entangled states", *j. j. franklin* **360**, 17(2023), doi:10.1016/j.jfranklin.2022.09.024

[11] Y. Yang, Z. Xu, A. del Campo, "Decoherence rate in random Lindblad dynamics", *Phys. Rev. Research* **6**, 023229 (2024), doi: 10.1103/PhysRevResearch.6.023229

[12] Y. Lu, et al., "Characterizing decoherence rates of a superconducting qubit by direct microwave scattering" *npj Quantum Information* **7**, 35(2021), doi: ; 10.1038/s41534-021-00367-5

Error Appendix

The value for the Rabi frequency was picked by judging where the maximum occured on fig.2. This introduced an error of $\pm\frac{0.02}{2\pi}MHz$. The error in calculating the fidelity was due to the small subset of input states that could be used. A model for the relationship in Fig.7 was not able to be found from which a deviation from the expected value could be calculated.

Scientific Summary for a General Audience

The classical world and the quantum world are governed by very different laws. So far we have been able to build classical computers to solve mathematical problems we have everyday. The innovation now is to build a computer that operates using quantum rules. The advantage of a quantum computer is that a system of quantum bits (qubits) can exist in a state which is a superposition of different values the qubit can take. This affords the quantum computer to have much more information stored for the same number of bits when compared to a classical computer where each bit has a definite value.
For the same number of bits, a quantum computer can have much more information stored due to this property. The current problem with quantum computers is that they don't give the correct output with 100% accuracy. Because of the behaviour of quantum objects, even small changes in the environment, such as a stray bit of light shining on it or surrounding particles, can change the value of the qubits. Understanding how much the qubits change in a certain environment can help mitigate these effects or even use them to give a correct computation.